Optimal Marine Aquaculture Zone analysis on the US West Coast

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Summary

In this project our goal was to build a function that could take a species' minimum and maximum optimal growing temperature, and minimum and maximum optimal growing depth, to find which Exclusive Economic Zones (EEZ) on the West Coast of the United States were best suited to developing it in marine aquaculture. We specifically took analyzed oysters (several species) and the Japanese Carpet Shell (Ruditapes philippinarum)

Initializing (loading libraries and data)

```
# Load libraries in
library(sf)
library(terra)
library(tidyverse)
library(tmap)
library(here)
library(kableExtra)
```

Data used in this analysis:

- `sst_files` - Sea surface temperature data from the years 2008 to 2012. Provided by NOAA.
- `wc_regions_clean.shp` - Exclusive economic zone marine boundary data for the west coast of the U.S. Provided by Marineregions.org
- `depth.tif` - Ocean Depth/Bathymetry data. From the general Bathymetric Chart of the Oceans.

```r
# Load data and preliminary checks
## Load in shapefile for the West Coast EEZ vector
wc_eez <- read_sf(here("data", "wc_regions_clean.shp"))

## Load in bathymetry raster
depth <- rast(here("data", "depth.tif"))

## Load in sea surface temp (sst) raster, combine files
sst_files <- list.files(path = here("data"), pattern = "sst", full.names = TRUE)
sst_stack <- rast(sst_files)

## Confirm all five years loaded as separate layers
if (nlyr(sst_stack) != 5) {
  warning(
    "SST stack does not contain 5 layers; check that all year files loaded"
  )
}
```

Verify coordinate reference systems

```r
## Check whether the CRS's match
if (st_crs(wc_eez) != st_crs(depth)) {
  warning("EEZ and depth CRS do not match; transformation required")
}

if (st_crs(depth) != st_crs(sst_stack)) {
  warning("Depth and SST CRS do not match; transformation required")
}
```

Process data

```r
# Process data
## combine all the layers into one by taking the average
sst_avg <- app(sst_stack, fun = mean)

## converting values to celcius from the original kelvin
sst_avg_C <- sst_avg - 273.15
```

```r
## crop depth raster to match the extent of the sst raster
depth_crop <- crop(depth, sst_avg_C)

## resample depth (higher res) with bilinear method
depth_resamp <- resample(depth_crop, y = sst_avg_C, method = "bilinear")

## make crs labels match
crs(sst_avg_C) <- crs(depth_resamp)
```

```r
## check depth_resamp and sst_avg_C on all metrics (resolution, extent, crs, and whether its
if (!all(res(sst_avg_C) == res(depth_resamp))) {
  warning("Resolutions do not match after resampling")
}

if (ext(sst_avg_C) != ext(depth_resamp)) {
  warning("Extents do not match after resampling")
}

if (st_crs(sst_avg_C) != st_crs(depth_resamp)) {
  warning("Coordinate reference systems do not match after resampling")
}

aligned_stack <- c(sst_avg_C, depth_resamp)
```

Find suitable locations

```r
# Find suitable locations

## reclassify sst and depth for oyster
rcl_oyster_sst <- matrix(
  c(-Inf, 11, 0, 11, 30, 1, 30, Inf, 0),
  ncol = 3,
  byrow = TRUE
)

rcl_oyster_depth <- matrix(
  c(-Inf, -70, 0, -70, 0, 1, 0, Inf, 0),
  ncol = 3,
  byrow = TRUE
```

```
)

## apply matrices
sst_reclass <- classify(sst_avg_C, rcl = rcl_oyster_sst)
depth_reclass <- classify(depth_resamp, rcl = rcl_oyster_depth)

## confirm reclassification produced only 0s and 1s
if (!all(unique(values(sst_reclass, na.rm = TRUE)) %in% c(0, 1))) {
  warning("SST reclassification produced values other than 0 and 1")
}

# stack reclassed rasters
sst_depth_stack <- c(sst_reclass, depth_reclass)

# define layer multiplication function
mult <- function(lay1, lay2) {
  lay1 * lay2
}

# apply the function
suitable_rast <- lapp(sst_depth_stack, fun = mult)
```

Determine the most suitable EEZ

```
# Determine the most suitable EEZ

## Area of grid cells
area_rast <- cellSize(suitable_rast, unit = "km")

## multiply rasters to get areas of suitable cells
suitable_area_rast <- suitable_rast * area_rast

## Rasterize eez
eez_raster <- rasterize(wc_eez, suitable_area_rast, field = "rgn_id")

## find which zone has the most suitable cells by area
most_suitable <- zonal(
  suitable_area_rast,
  eez_raster,
```

```
  fun = "sum",
  na.rm = TRUE
)

## Left Join
most_suitable_oyster <- left_join(wc_eez, most_suitable, by = "rgn_id")
most_suitable_oyster <- rename(most_suitable_oyster, suitable_area_km2 = lyr1)
```

```
## table of results, with suitable area also expressed as a share of each EEZ
most_suitable_oyster %>%
  st_drop_geometry() %>%
  select(rgn, suitable_area_km2, area_km2) %>%
  mutate(
    pct_of_eez = round(100 * suitable_area_km2 / area_km2, 2),
    suitable_area_km2 = round(suitable_area_km2, 1),
    area_km2 = round(area_km2, 1)
  ) %>%
  arrange(desc(suitable_area_km2)) %>%
  kable(
    col.names = c(
      "EEZ region",
      "Suitable area (sq km)",
      "Total EEZ area (sq km)",
      "Percent of EEZ suitable"
    ),
    caption = "Suitable aquaculture area for oysters by West Coast EEZ"
  ) %>%
  kable_styling(bootstrap_options = "striped", latex_options = "hold_position")
```

Table 1: Suitable aquaculture area for oysters by West Coast EEZ

EEZ region	Suitable area (sq km)	Total EEZ area (sq km)	Percent of EEZ suitable
Central California	3656.8	202738.3	1.80
Southern California	3062.2	206860.8	1.48
Washington	2435.9	66898.3	3.64
Oregon	1028.9	179994.1	0.57
Northern California	194.1	164378.8	0.12

Central California rankes the highest in terms of raw suitable area with 3,656.8 square kilometers. However in terms of a percentage of suitable area to total area Washington is the highest at 3.64 percent.

Generalizing the workflow

This function takes the following arguments:

- `min_temp` - Minimum optimal growth temperature
- `max_temp` - Maximum optimal growth temperature
- `min_depth` - Minimum optimal growth depth
- `max_depth` - Maximum optimal growth depth
- `species_name` - Name of the species

The function outputs:

- A map of the different zones with the hue of blue representing different amounts of suitable area.

```
# Function creation
suitability_map <- function(
  min_temp,
  max_temp,
  min_depth,
  max_depth,
  species_name
) {
  # Check that user did not accidentally flip min and max temp
  if (min_temp >= max_temp) {
    stop("min_temp must be less than max_temp")
  }

  # Check that user did not accidentally flip min and max depth
  if (min_depth >= max_depth) {
    stop(
      "min_depth must be less than max_depth (both are negative values below sea level)"
    )
  }

  # Find suitable locations

  ## reclassify sst and depth
  rcl_sst <- matrix(
    c(-Inf, min_temp, 0, min_temp, max_temp, 1, max_temp, Inf, 0),
    ncol = 3,
    byrow = TRUE
  )
```

```r
rcl_depth <- matrix(
  c(-Inf, min_depth, 0, min_depth, max_depth, 1, max_depth, Inf, 0),
  ncol = 3,
  byrow = TRUE
)

## apply matrices
sst_reclass <- classify(sst_avg_C, rcl = rcl_sst)
depth_reclass <- classify(depth_resamp, rcl = rcl_depth)

# apply the function
suitable_rast <- lapp(c(sst_reclass, depth_reclass), fun = mult)

# Determine the most suitable EEZ

## multiply rasters to get areas of suitable cells
suitable_area_rast <- suitable_rast * area_rast

## Rasterize eez
eez_raster <- rasterize(wc_eez, suitable_area_rast, field = "rgn_id")

## find which zone has the most suitable cells by area
most_suitable <- zonal(
  suitable_area_rast,
  eez_raster,
  fun = "sum",
  na.rm = TRUE
)

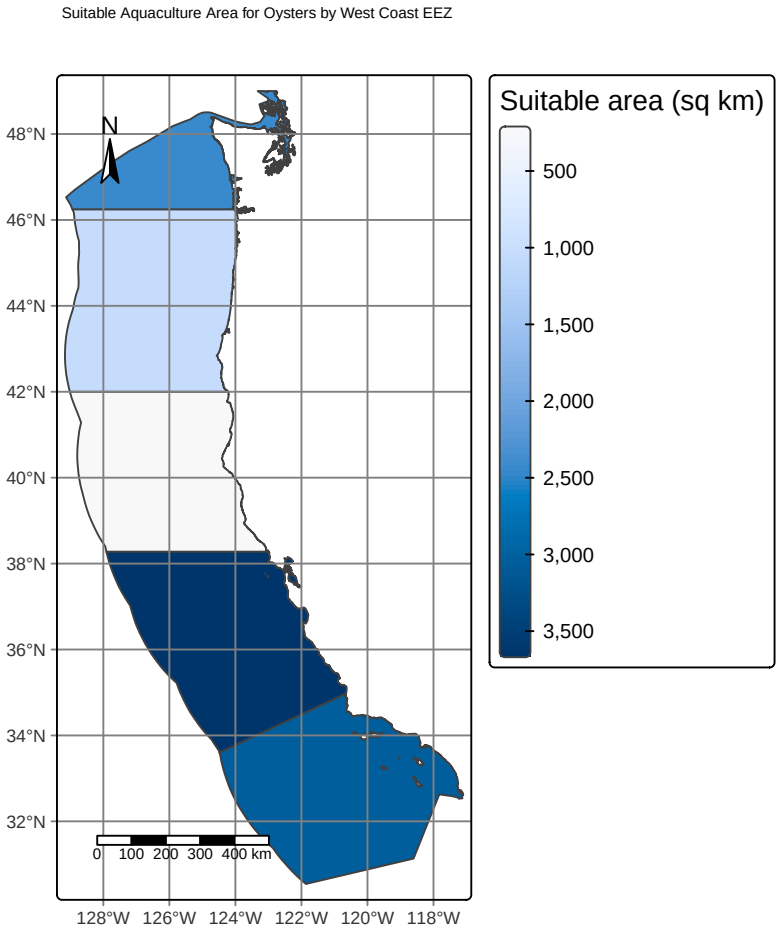
## Left Join
eez_suitable <- left_join(wc_eez, most_suitable, by = "rgn_id")
eez_suitable <- rename(eez_suitable, suitable_area_km2 = lyr1)

# Plot
## continuous scale rather than classed intervals, since the distribution of
## suitable area varies by species and binning produced degenerate breaks
tm_shape(eez_suitable) +
  tm_polygons(
    fill = "suitable_area_km2",
    fill.scale = tm_scale_continuous(),
    fill.legend = tm_legend(title = "Suitable area (sq km)")
  ) +
```

```
    tm_graticules() +
    tm_compass(position = c("left", "top"), size = 1.5) +
    tm_scalebar(position = c("left", "bottom"), text.size = 0.5) +
    tm_title(
      text = paste(
        "Suitable Aquaculture Area for",
        species_name,
        "by West Coast EEZ"
      )
    ) +
    tm_layout(legend.outside = TRUE, legend.outside.size = 0.2)
}
```

Oysters

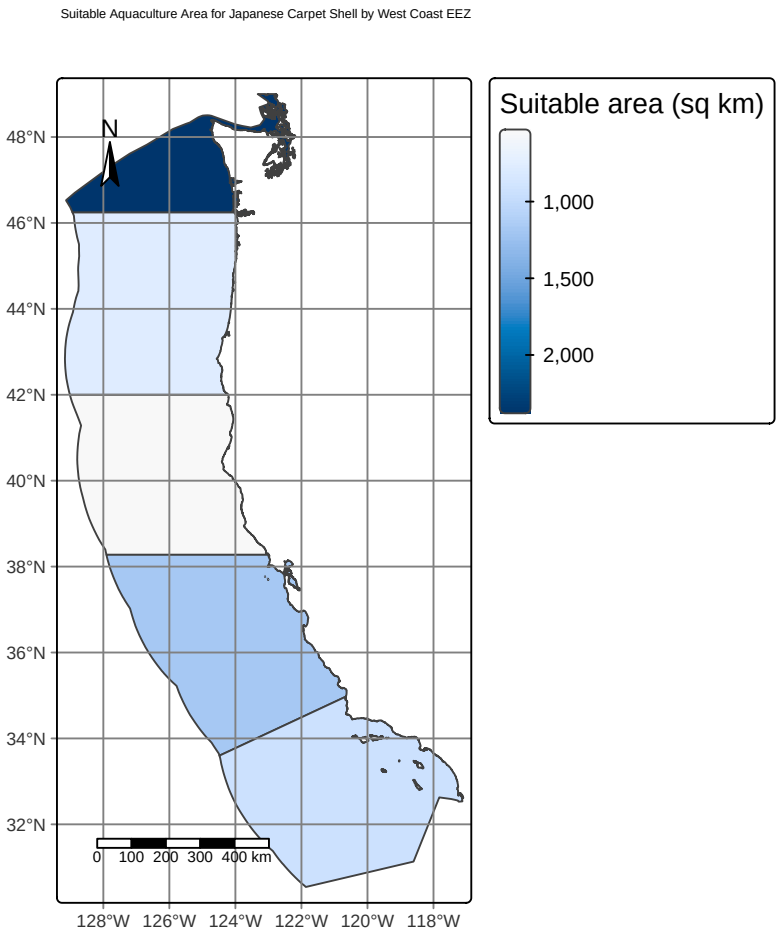
```
# Test function works with oyster
suitability_map(11, 30, -70, 0, "Oysters")
```



Japanese carpet shell (*Ruditapes philippinarum*)

I chose Japanese carpet shell as my species of choice.

```
# Test function works with species of choice (manila clam)
suitability_map(0, 23, -30, 0, "Japanese Carpet Shell")
```



According to our method Washington is the most suitable economic exclusive zone for Japanese Carpet Shell

Data citations

- NOAA Coral Reef Watch 5km Daily Global Satellite Sea Surface Temperature Anomaly v3.1
- GEBCO Compilation Group (2022) GEBCO_2022 Grid, doi:10.5285/e0f0bb80-ab44-2739-e053-6c86abc0289c
- Marineregions.org, Exclusive Economic Zones
- FAO (2009) Cultured aquatic species fact sheet, Ruditapes philippinarum, text by Goulletquer, P.
- SeaLifeBase, Ruditapes philippinarum species summary, citing SAUP Database 2006